DAY 5 — Exploratory Data Analysis (EDA) & Visualization Basics

Day 5 was one of the most exciting days — we entered the world of **EDA (Exploratory Data Analysis)**. EDA helps us "understand" the dataset before building any model. It allows us to observe trends, relationships, patterns, and anomalies.

🔹 Why EDA is Important?

EDA helps answer:

- What does the dataset contain?
- What trends are visible?
- Are there outliers?
- Are columns correlated?
- What factors influence the target variable?

EDA transforms raw numbers into insights.

🔸 1. Dataset Exploration

We used:

- df.head()
- df.tail()
- df.info()
- df.describe()
- df.nunique()
- df.value_counts()

This gave a clear overview of numerical vs categorical features, ranges, distributions, and unique values.

🔸 2. Univariate Analysis

Studying one column at a time:

- Histogram
- Box plot
- Bar chart

These helped identify:

- Data distribution

- Outliers
- Frequency of categories

🔸 3. Bivariate Analysis

Studying relationships between two variables:

- Line plots
- Scatter plots
- Heatmaps
- Correlation matrix

We used:

df.corr()

plt.scatter()

sns.heatmap()

Correlation values helped interpret which features impact stock price movement.

🔸 4. Visualizations with Matplotlib & Seaborn

We learned to plot:

- Line charts
- Histograms
- Bar plots
- Count plots
- Box plots
- Heatmaps

Visualization is crucial because **pictures reveal what tables cannot**.

🔸 5. Identifying Outliers

Using boxplots and IQR (Interquartile Range):

IQR = Q3 - Q1

Upper = Q3 + 1.5*IQR

Lower = Q1 - 1.5*IQR

Outliers affect model predictions and must be handled.

Key Takeaways from Day 5

- EDA helps understand the dataset deeply
- Visualization simplifies complex data
- Finding correlations helps choose good features
- EDA sets the direction for ML model accuracy

This day made data feel like a story — full of patterns waiting to be discovered.